

# SILICON DREAMS

## Study Guide

MODULE 3 · WEEK 3 · CM-HW-101-M3

SG-M3-06 pairs with TB-M3-06

## LibreLane and DRC Fixes

*Read a DRC report, classify the top five violation types, fix them in the correct layer of the flow.*

### Learning objectives

- Navigate runs/<tag>/reports/signoff/drc.rpt and read Magic's error format.
- Classify every DRC error into one of five buckets: met1 spacing, antenna ratio, tap-cell spacing, power via enclosure, off-grid pins.
- Know which config knob or RTL change fixes each bucket — NOT all DRC errors are fixed the same way.
- Achieve zero DRC on the full design.

### Step 1 · Locate the DRC report

```
cd runs/<latest-tag>/reports/signoff
less drc.rpt
```

```
# Format:
# [CATEGORY] line: <error message>
# cell: <offending cell>
# box: ( xmin ymin xmax ymax )
# layer: met1 / met2 / tap / etc.
```

### Step 2 · The five buckets

#### met1 spacing

Two metal-1 shapes violate minimum spacing. Usually caused by high PL\_TARGET\_DENSITY\_PCT. Lower from 0.60 to 0.55, i In some versions of the tool, density is expressed as a decimal: 0.55 = 55%, 0.60 = 60% re-run.

|                            |                                                                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>antenna ratio</b>       | A long metal wire acts as an antenna during fab, can damage gates. Fix: insert antenna diodes (`DIODE_INSERTION_STRATEGY=3` in config.json) or break the net with a buffer.                                    |
| <b>tap cell spacing</b>    | sky130 requires a tap cell every 15 $\mu\text{m}$ . If you see this, your floorplan has a gap. Set `FP_TAP_DISTANCE=14` to force denser tap insertion.                                                         |
| <b>power via enclosure</b> | The via from met1 to met4 in the power straps isn't fully enclosed by the surrounding metal. Fix: bump `FP_PDN_ENABLE_RAILS_POWER_GRID` and widen the strap — usually a floorplan config knob, NOT an RTL fix. |
| <b>off-grid pins</b>       | Your pad-ring pin is not on the manufacturing grid. Check `pin_order.cfg` — x/y must be multiples of 0.005 $\mu\text{m}$ .                                                                                     |

### Step 3 · Worked example — met1 spacing

Typical snippet of drc.rpt for this category:

```
[DRC-1] line 443: Metal1 to Metal1 spacing violation (min 0.14  $\mu\text{m}$ )
  cell: _24837_
  box: (12.345 34.567 12.485 34.707)
  layer: met1
  severity: ERROR
```

The fix:

```
# Edit librelane/config.json
"PL_TARGET_DENSITY_PCT": 0.55,    # was 0.60
"ROUTING_OPT_ITERS": 32,          # was 16 — more passes give router more room

# Re-run
./scripts/run-librelane.sh

# Diff the report line counts
wc -l runs/first-run/reports/signoff/drc.rpt \
    runs/drc-fix-1/reports/signoff/drc.rpt
```

### Step 4 · Worked example — antenna

```
# config.json
"DIODE_INSERTION_STRATEGY": 3,
"GRT_REPAIR_ANTENNAS": 1
```

Strategy 3 inserts antenna diodes on every high-antenna-ratio net. If you're power-tight, try strategy 2 first — it is less invasive but won't fix all cases.

### Step 5 · Iterate

DRC fixing is iterative. Each pass:

1. Read the top error category by count.
2. Apply the canonical fix for that category (see table above).
3. Re-run LibreLane with a new tag.
4. Compare drc.rpt line count. Expect exponential decay.

Most designs in this cohort reach zero DRC in 3-4 iterations. If you're on iteration 6 and still have >50 errors, you likely have a structural floorplan problem — escalate in Discord #m3-drc.

## Step 6 · CI integration

``.github/workflows/drc.yml`` POSTs every failing DRC report to the ChipFoundry grader. It does NOT fail the PR on DRC errors (those are noisy during iteration), but it DOES fail the final-sign-off tag ``v1.0.0-final``.

```
# Tag the run when you believe it's clean
git tag v1.0.0-final
git push --tags

# CI picks up the tag and runs:
# 1. DRC must be 0 errors
# 2. LVS must be MATCH
# 3. Timing WNS must be >= 0 (worst corner)
# Any fail → tag is rejected, you must re-tag as v1.0.1-final etc.
```

## Troubleshooting

|                                                |                                                                                                                            |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>DRC count oscillates between runs</b>       | LibreLane is non-deterministic in placement. Pin the random seed in config.json:<br>"PL_RANDOM_INITIAL_PLACEMENT_SEED": 42 |
| <b>Same error after fix</b>                    | Your fix was in the wrong config key. Double-check the spelling — LibreLane doesn't error on unknown keys.                 |
| <b>New category appears after fix</b>          | Normal. Fixing one category can shift placement and expose another. Keep iterating.                                        |
| <b>drc.rpt is empty but CI says DRC failed</b> | Check drc.klayout.xml — KLayout runs a second DRC pass with different rules. Both must be clean.                           |

## Reflection

- Record your DRC error count per iteration. Plot it — the curve should be roughly geometric.
- Which bucket gave you the most trouble? That's the one to write up in your escape log. Expect 'met1 spacing' for most learners.